

Received 23 March 2023, accepted 12 April 2023, date of publication 14 April 2023, date of current version 19 April 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3267409

## RESEARCH ARTICLE

# Violence Detection Enhancement by Involving Convolutional Block Attention Modules Into Various Deep Learning Architectures: Comprehensive Case Study for UBI-Fights Dataset

MAHMOUD ABDELKADER BASHERY ABBASS<sup>1,2</sup>, (Member, IEEE),  
AND HYUN-SOO KANG<sup>3</sup>, (Member, IEEE)

<sup>1</sup>Department of Information and Communication Engineering, School of Electrical and Computer Engineering, Chungbuk National University, Cheongju-si 28644, South Korea

<sup>2</sup>Department of Mechanical Power Engineering, Helwan University, Cairo 11772, Egypt

<sup>3</sup>Department of Information and Communication Engineering, School of Electrical and Computer Engineering, Chungbuk National University, Cheongju-si 28644, South Korea

Corresponding author: Hyun-Soo Kang (hskang@cbnu.ac.kr)

This work was supported in part by the Ministry of Science and ICT (MSIT), South Korea, under the Grand Information Technology Research Center Support Program Supervised by the Institute for Information & Communications Technology Planning & Evaluation (IITP) under Grant IITP-2023-2020-0-01462; and in part by the Research Projects of Development of Automatic Screening and Hybrid Detection System for Hazardous Material Detecting in Port Container funded by the Ministry of Oceans and Fisheries.

**ABSTRACT** Violence detection in surveillance videos is a complicated task, due to the requirements of extracting the spatio-temporal features in different videos environment, and various video perspective cases. Hereby, in this paper, different architectures are proposed to perform this task in high performance, by using the UBI-Fights dataset as a comprehensive case study. The proposed architectures are based on involving the Convolutional Block Attention Modules (CBAM) with other simple layers (e.g., ConvLSTM2D or Conv2D&LSTM). In addition, using the Categorical Focal Loss (CFL) as a loss function during various architecture training, to increase the focus on the most important features. To evaluate the proposed architectures, the performance metrics like Area Under the Curve (AUC), and Equal Error Rate (EER); are mainly used, to declare the architecture's ability to identify the violence correctly, with low interaction value between classes. The performance results declare the ability of the proposed architectures, to achieve higher results than the state-of-the-art techniques. For example, the Conv2D&LSTM-based architecture gets an AUC value of 0.9493, and an EER value of 0.0507; which outperforms most of the other proposed ones, and the state-of-the-art performance.

**INDEX TERMS** Area under the curve metric, categorical focal loss, ConvLSTM2D, convolutional block attention module, equal error rate metric, spatio-temporal features, UBI-fights data, violence detection.

## NOMENCLATURE

*Symbol Description*

AUC Area Under the Curve  
CBAM Convolutional Block Attention Module  
CCE Categorical Cross-Entropy  
CFL Categorical Focal Loss

CNN Convolutional Neural Network  
Conv-LSTM Convolutional Long Short-Term Memory  
CV Computer Vision  
EER Equal Error Rate  
FAR False Acceptance Rate  
FRR False Rejection Rate  
GMM Gaussian Mixture Model  
MIL Multiple Instance Learning  
RNN Recurrent Neural Network

The associate editor coordinating the review of this manuscript and approving it for publication was Inês Domingues<sup>id</sup>.

## I. INTRODUCTION

Human behavior detection, in general, and violence detection in specific have recently become a significant area of Computer Vision (CV) study due to the ongoing increase in abnormal behaviors in different places. Furthermore, due to the environment complex (i.e., social interactions) and the difficulties in extracting a specific characteristic that connects with a specific occurrence, violence detection is one of the most difficult challenges in CV [1]. In other words, the detection of a violent situation accurately needs two main feature extraction methods: 1) Spatial or shape feature extraction, and 2) Temporal or time features extraction. The spatial features represent the relationships or interactions between single-frame pixels, but it is not enough to identify the violence. Because, the spatial features only can't identify the people's behaviors through time (e.g., if the player touching in an American football match is for hitting each other's, or for playing together). Hereby, the importance of the second feature extraction method (i.e., temporal features extractions) comes here, which this features type is able to catch the interaction between sequential frames, to get the nature of people's behavior through time [2], [3].

Meanwhile, the most popular work in violence detection, is done based on surveillance videos, to extract spatio-temporal features, that help in identifying violence cases clearly [4], [5], [6], [7], [8], [9], [10]. Hereby, this paper proposed different architectures based on extracting the spatio-temporal features by using different techniques (e.g., 3D Convolutional Neural Network (CNN) [11], [12], [13], Convolutional Long Short-Term Memory (Conv-LSTM) networks [5], [7], [14], [15], integrating transfer learning with LSTM [16] or Conv-LSTM [7]), and adding a combination of attention modules (i.e., channel attention, and spatial attention) into the architectures to increase the overall classification performance [17]. In the next subsections, a review for the related work of violence detection based on the UBI-Fighting data, is presented. This comparison declares the innovation and novelty of the proposed work, with respect to the previous case studies on the same data type (i.e., UBI-Fights).

### A. RELATED WORK

Recently, there are many significant work for the violence detection, has been done based on the UBI-Fights video data. For example, Bruno Manuel Degardin in [18] proposes a complicated iterative learning framework based on Bayesian filtration for the instances of unlabeled data, to create weak/self-supervised learning. In addition, the author used the random forest algorithm that containing fifty decision trees, to enhance the overall performance of three models by using late decision fusion ensemble technique. The conclusion stated that the performance of this framework on the UBI-fights data; is 0.819 for the Area Under the Curve (AUC) metric, and 0.284 for Equal Error Rate (EER) metric.

Furthermore, Weijun Tan et al. in [19] made a comprehensive violence detection studies by implementing different

techniques (e.g., the Inception 3D model (I3D), a special ResNet-type architecture called (R(2+1)D) based on making a separate extraction of temporal and spatial features by using a 3D convolutions, and an attention mechanism of Bi-directional Encoder Representations from Transformers (BERT) [20]) on the UBI-Fights data. One of these studies, the authors integrated the BERT attention mechanism with R(2+1)D architecture, to enhance the overall AUC classification metric to 0.915.

Also, by using the UBI-Fights data, Mohammed Alarfaj et al. [21] tried to achieve high classification performance by increasing the preprocessing steps (e.g., moving object detection, region shrinking technique, and extracting objects' temporal features), before the XG-Boost classifier. The authors implemented a preprocessing data flow based on many steps, starting with using a combination of two algorithms (i.e., Gaussian Mixture Model (GMM), and Kalman filter), to detect the moving objects. Then, they invent a shadow segmentation and occluded human detection method by using a region-shrinking technique. After that, a temporal features extraction (e.g., velocity, and optical flow) was done by using a Fuzzy c-mean algorithm based on the Hough transformation technique. The final preprocessing step was the computation cost reduction by implementing a Gray Wolf Optimization (GWO) on the extracted feature level. The conclusion stated that the implementation of the overall technique on the UBI-Fights data, got an AUC value of 0.769, and an EER value of 0.427.

Similarly, Zhenting Qi et al. in [8] tried to perform a complicated framework for violence detection and used the UBI-Fights data to evaluate the framework. The framework was based on making weak supervision during training; by implementing a two stages training for a violence detection model. This model contained a feature extraction technique as a preprocessing head (called action-aware because of the special attention to people's actions was given during extracting features), and an anomaly score generator as a tail (this score is based on the spatio-temporal extracted features from the action-aware feature extractor). The first training stage for this model is called score generator training, because it focused on anomaly score generator training by using Multiple Instance Learning (MIL) loss, to maximize the classification margin between violence and non-violent videos. Then, the second stage of training, is a type of self-learning strategy, that refines the score generator by putting soft Pseudo Labels for the violence videos, to avoid overconfidence; and using cross entropy as a loss function. The conclusion mentioned the performance of this framework on the UBI-Fights data; as 0.931 for ACC, and 0.085 for EER.

### B. CONTRIBUTIONS AND PROPOSED APPROACH

The main approach through this paper and its contribution also, are as follows:

- A full demonstration of the nature of violence detection of surveillance videos task, with a declaration for

**TABLE 1.** Review for the most recent violence detection papers and comparison in terms of AUC and EER.

| Reference                    | Technique                                                                                                                                                                 | AUC   | EER   | Comment                                                                                                                                                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bruno Manuel Degardin [18]   | Iterative learning framework based on Bayesian filtration                                                                                                                 | 0.819 | 0.284 | Complex framework and need huge time to make iterative strategy                                                                                                                                                                                              |
| Weijun Tan et al. [19]       | Combine R(2+1)D architecture with BERT attention                                                                                                                          | 0.915 | —     | Huge architecture, due to the 3D convolutional that need high computational resources and consume time                                                                                                                                                       |
| Mohammed Alarfaj et al. [21] | XG-Boost classifier with preprocessing steps (e.g., moving object detection, region shrinking technique, extracting objects temporal features, and features optimization) | 0.769 | 0.427 | Complex preprocessing steps, and the overall technique is not suitable for heavy density crowded environments. Because of, the human silhouette extraction technique                                                                                         |
| Zhenting Qi et al. [8]       | Weakly supervised training of two stage framework for feature extractor and anomaly score generator model                                                                 | 0.931 | 0.085 | Complex framework because it needs two stages training, o it needs huge training time. In addition, the first training stage based on loss technique that weak in considering the temporal features between frames, and unable to deal with noise situations |

the challenges through performing solutions. In addition, a review of the most recent papers achieves high-performance results by using the UBI-Fights dataset.

- Proposing different architectures based on integrating the Convolutional Block Attention Modules (CBAM) with various layers such as ConvLSTM2D or Conv2d&LSTM layers; to catch the spatio-temporal features, and increase the focus on the important ones. Furthermore, the using for Categorical Focal Loss function (CFL) through the training, increases the focus on the important features, and overcomes the drawback of class imbalance data.
- Making two Comparisons to declare the significance of the proposed work results, by comparing the simply proposed architectures with other complicated ones; and also with respect to the state-of-the-art on the same data.

The rest of this paper contains six sections besides the introduction section, that organized as follows: the model architecture declaration with a full demonstration for each part, in section II; the explanation for the UBI-Fights data in section III; the details of the experiments that including the training methodology, evaluation metrics, and ablation study in section IV; the result and analysis of all experiments, and the comparison between the ablation study and state-of-the-art in section V; the brief discussion about the detailed results explanation in section VI; and finally, the conclusion and future work for all proposed work in section VII.

## II. MODEL ARCHITECTURE

To get the highest violence detection performance on the UBI-Fights data, different architectures are proposed, based on the main concept of detecting the spatio-temporal features of each video to classify it correctly. The proposed architectures are based on integrating state-of-the-art techniques such as Convolutional Block Attention Module (CBAM) of spatial attention and channel attention modules [17], into the traditional ways of catching spatio-temporal features (e.g., CNN+LSTM [6], and ConvLSTM [7], [9]); to increase the attention about the most violence features per video. In addition, due to the category imbalance of the UBI-Fights data, the categorical focal loss is used as a loss function [22].

In the next subsections, the main components of each architecture, are demonstrated. Also, a detailed explanation for the proposed architectures is explained, and visualized

### A. MAIN ARCHITECTURES COMPONENTS JUSTIFICATIONS

In this subsection, the main backbones or items that used in the proposed architectures, are clearly explained, with a full declaration for each justification and adaptation, to be suitable for the violence detection from surveillance videos case study.

#### 1) DENSENET CONVOLUTIONAL 2D

The DenseNet architecture is made up of 2D convolutional layers since it was intended to be used with frames. The dense blocks that serve as the framework for the network

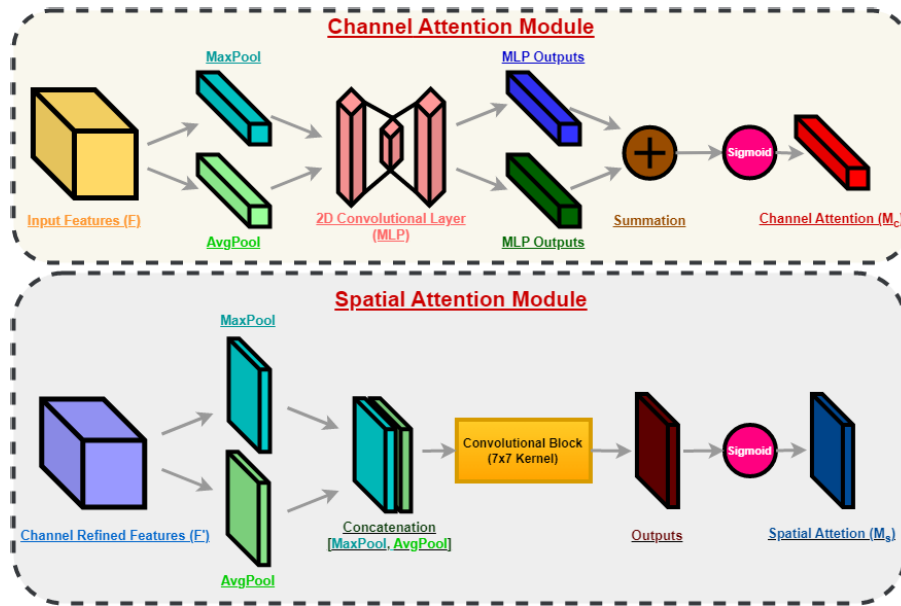


FIGURE 1. The CBAM sub-modules of channel and spatial attention.

architecture are where DenseNet gets its name; these blocks combine a layer's feature maps with all of its followers [23]. The main use for this architecture is the identification between each frame's pixels, to catch the spatial features between pixels in each frame perfectly. However, the violence detection of surveillance videos, is a 3D case study. Because, the videos contain the third dimension of frames changing with time. In other words, each frame in the video has a dimension of (height, width, channels), which the overall video has a dimension of (frames, height, width, channels). So, the justification for DenseNet architecture, is to be suitable for the 3D case study; by evolving the DenseNet 2D by a TimeDistribution layer to get the temporal features between video frames [6]. In addition, the other performance enhancing 2D layers type such as MaxPool2D; must be replaced by MaxPool3D. Similarly, for the transfer learning models like ResNet, VGG, and Mobilenet; the architectures are evolved by the TimeDistribution layer, to catch the spatio-temporal features through videos.

## 2) CONVOLUTIONAL LSTM 2D

Standard Recurrent Neural Networks (RNNs) have the ability to extract temporal features that change in the dimension of time, but has the drawback of losing spatial ones. To solve this issue, the convolutional layers are used in place of completely linked ones to increase the network's ability to get the spatio-temporal features. In other words, adding the convolution layers into the RNN enables the network to fully utilize the spatio-temporal information that results from the correlation between convolution and recurrent operations and allows the ConvLSTM layer to learn the spatio-temporal aspects of a video [5], [7], [9]. Furthermore, in some cases, we need to integrate the ConvLSTM layer inside a Bidirectional layer;

it is an additional computation cost, but is sometimes used to increase the network performance by analyzing frames sequences in forward and backward simultaneously [7].

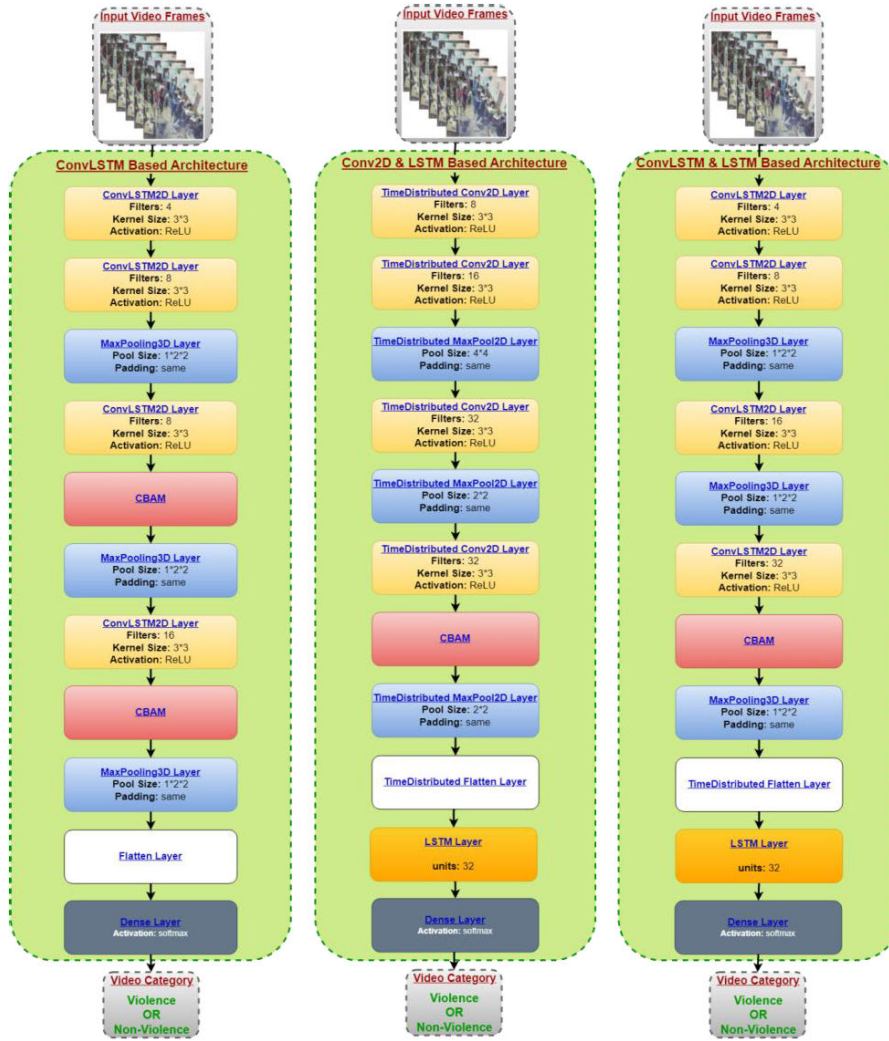
BiConvLSTM is an enhancement on ConvLSTM that allows to analysis of sequences forward and backward in time simultaneously. A BiConvLSTM layer can access information in both directions of a video's timeline. In this way, a better overall understanding of the video is achieved.

## 3) CONVOLUTIONAL BLOCK ATTENTION MODULE

Through the training process, CNN may focus on unnecessary features. Herby, CNN is taught and focused more on vital information using attention modules than on unhelpful previous knowledge. In the case of violence detection, useful information is the nature of interaction dynamics between people that we want to classify and localize in an image. Hereby, the Convolutional Block Attention Module (CBAM) is proposed in [17] and its concept is used in [24], [25], and [26]; that consists of two main sub-modules (as shown in Figure 1): 1) Channel attention module, and 2) Spatial attention module. As illustrated in Figure 1, the input features are input to the channel attention, in the shape of  $F \in C \times H \times W$ . then, the CBAM separates the input features map ( $F$ ) sequentially into a 1D channel attention map as  $M_c \in C \times 1 \times 1$ , and a 2D spatial attention map  $M_s \in 1 \times H \times W$ .

For the channel attention module, it is represented by applying the sigmoid function on the output of a simple 2D convolutional layer called (MLP); which takes the output of AvgPool and Maxpool, as shown in equation (1). Furthermore, the spatial attention module constructs new two features maps by using the AvgPool and MaxPool; then makes concatenation for those features maps, to pass them into a convolutional block with  $7 \times 7$  kernel; and finally applies





**FIGURE 2.** The proposed simple architectures in terms of CBAM integration with ConvLSTM, Conv2D, and LSTM layers.

the sigmoid function as shown in equation (2). The general attention process can be encapsulated as two equations that are implemented sequentially on the input feature map ( $F$ ); as shown in equation (1), and equation (2); to get the final attention map ( $F''$ ). This sequential process is done based on element-wise multiplication, which has a symbol of ( $\otimes$ ) in equations (3) and (4).

$$M_c(F) = \text{sigmoid}(f^{7 \times 7} ([\text{AvgPool}(F); \text{MaxPool}(F)])) \quad (1)$$

$$M_c(F) = \text{sigmoid}(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \quad (2)$$

$$F' = M_c(F) \otimes F \quad (3)$$

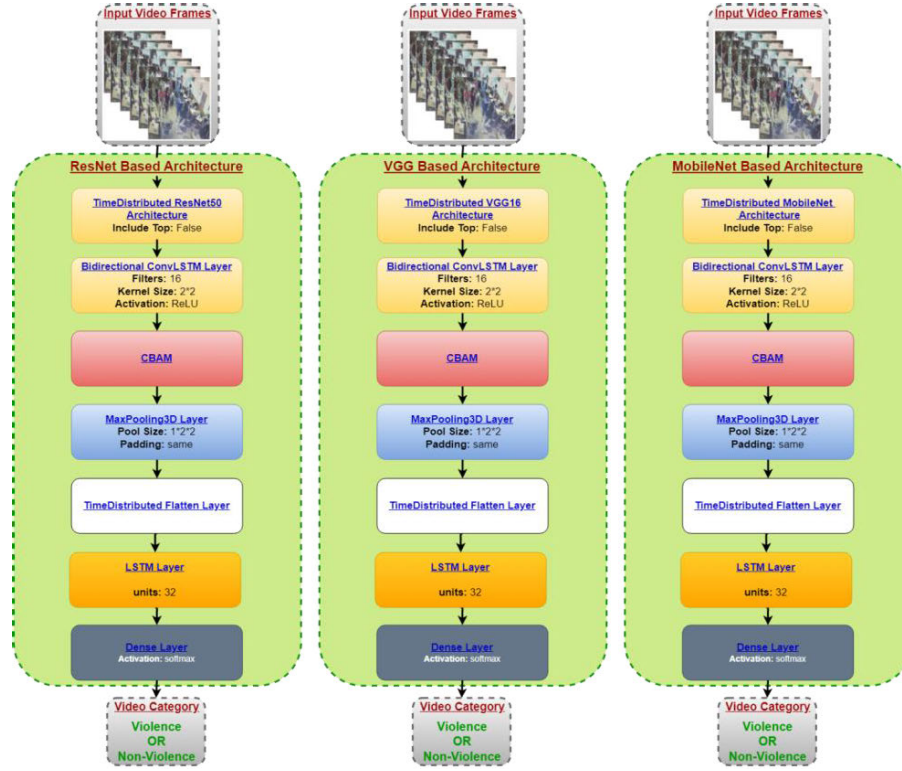
$$F'' = M_c(F') \otimes F' \quad (4)$$

#### 4) CATEGORICAL FOCAL LOSS

The violence detection of surveillance videos is known as a classification problem with two category cases. The most

well-known loss functions in such case study, such as the Categorical Cross-Entropy (CCE); are not recommended in a such case study on the UBI-Fights data. Due to, the class imbalance for the used dataset, using such loss functions is not the right choice; because the bias is transmitted as a result of class imbalance. The weights will update such that the model is increasingly confident in forecasting the majority class while placing less focus on the minority classes as a result of the majority class instances dominating the loss function and gradient descent [22], [27].

To solve this issue, Categorical Focal Loss (CFL) is used; because it has the ability to overcome such problem cases of imbalanced class datasets. The categorical focal loss function, is based on two main techniques: 1) Down weighting with focusing parameter ( $\gamma$ ), and 2) Balancing variant with weighing factor ( $\alpha$ ). Those techniques, are methods that highlight the low number of sampled categories more by minimizing the impact of the higher category samples on the loss function. This adjustment process is done by reducing the



**FIGURE 3.** The proposed transfer learning based architectures in terms of CBAM integration with ResNet50, VGG16, and MobileNet architectures.

modulating factor for the correctly classified classes; and increasing it for the misclassified classes, to adjust model weights by using the loss value as shown in equation (5). As recommended in [22], the selected values for this technique are 2 for ( $\gamma$ ), and 0.25 for  $\alpha$ .

$$CFL = - \sum_{i=1}^{i=n} \alpha_i (i - (Probability)_i)^\gamma \log((Probability)_i) \quad (5)$$

## B. THE PROPOSED ARCHITECTURES

After knowing the main basic blocks for the proposed work; the detailed proposed architectures, are discussed in this subsection. As shown in Figure 2 and Figure 3, the proposed architectures are divided into two main categories. The first one is the simple architectures that are created from scratch (Figure 2); while the other category is the transfer learning enhanced architectures. Both categories contain the CBAM somehow, to increase the overall performance for each one. Each one will be explained clearly, in the next paragraphs.

### 1) CONVLSTM-BASED ARCHITECTURE

This architecture starts with a combination of two ConvLSTM2D layers and one MaxPooling3D layer (this combination is used as a feature extraction combination). For the initial two layers, they have similar specifications of  $3 \times 3$  kernel size, and ReLU activation function; but the number of filters is 4, and 8, respectively; and the

MaxPooling3D layer with a pool size of  $1 \times 2 \times 2$ . Then, another type of combination is added, to extract much more spatio-temporal features and increase the attention to the most important extracted features. This combination includes one ConvLSTM2D layer with 8 filters,  $3 \times 3$  kernel size, and ReLU activation; the CBAM; and MaxPooling3D layer with the same specification as the similar previous one (i.e., pool size of  $1 \times 2 \times 2$ ). Another, similar combination (i.e., ConvLSTM2D layer, CBAM; and MaxPooling3D layer) is added to the architecture, but the number of filters for the ConvLSTM2D layer increases again to 16. Finally, the final two layers are the Flatten, and Dense layers with Softmax activation; to make the final decision of video category (i.e., violence or non-violence).

### 2) CONV2D AND LSTM-BASED ARCHITECTURE

In this one, the concept of extracting the spatio-temporal features is done, by using the TimeDistributed layer as a cover for all 2D layers, to extract the 3D features by using the 2D-based features extractors. This architecture is started by two feature extractor combinations; the first one contains two Conv2D layers, which are involved inside a TimeDistributed layer, with common specifications of  $3 \times 3$  kernel size, and ReLU activation function; but the first Conv2D layer has 8 filters, and the second one has 16 filters; and the final layer in this combination is a MaxPool2D layer, that included inside a TimeDistributed layer too, with a pool size of  $4 \times 4$ .

The second combination is much more simply, because it has just one conv2D layer with 32 filters,  $3 \times 3$  kernel size, and ReLU activation function; and the other one is a MaxPool2D layer with  $2 \times 2$  pool size; both of those layers are included in TimeDistributed layers too. Then, a different combination type is added, to extract more spatio-temporal features and increase the attention to the most important ones; this combination has a Conv2D layer, included in a TimeDistributed layer, that is similar to the last one (i.e., with 32 filters,  $3 \times 3$  kernel size, and ReLU activation function); is followed by a CBAM; and finally a MaxPool2D layer with  $2 \times 2$  pool size, inside a TimeDistributed one. Finally, three layers are added to take the final decision for the video type; the Flatten layer is covered by a TimeDistributed layer; then an LSTM layer with 32 units; and the final layer is the Dense layer with two nodes and Softmax activation function to identify the video.

### 3) CONVLSTM AND LSTM-BASED ARCHITECTURE

This architecture represents another shape of the ConvLSTM-based architecture, that doesn't achieve high AUC value, by comparing it to the state-of-the-art values. This target is done by making two spatio-temporal features extractors; the first is similar to the first one in ConvLSTM-based architectures (i.e., the first combination has two ConvLSTM2D layers with similar specifications of  $3 \times 3$  kernel size, and ReLU activation function, but the filters numbers are 4, and 8 for the first and second ones, respectively; and one MaxPooling3D layer with  $1 \times 2 \times 2$  pool size); the second combination has just one ConvLSTM2D layer (16 filters,  $3 \times 3$  kernel size, and ReLU activation function), and one MaxPooling3D layer with  $1 \times 2 \times 2$  pool size. Then, another attention-based combination is added, by adding a new ConvLSTM2D layer with 32 filters,  $3 \times 3$  kernel size, and ReLU activation function; then the CBAM is added; followed by a MaxPooling3D layer with  $1 \times 2 \times 2$  pool size. Finally, the same last three layers in the Conv2D&LSTM-based architecture (i.e., TimeDistributed layer including Flatten one, LSTM layer with 32 units, and Dense layer with Softmax activation), are added to this one to identify the video type (i.e., violence or non-violence).

### 4) TRANSFER-LEARNING-BASED ARCHITECTURES

There are an additional three architectures that are based on transfer learning architectures, to be used as spatio-temporal feature extractors, by involving them in a TimeDistributed layer. Those three architectures are similar in everything, except the first part which represents the feature extraction part. This first part contains three well-known architectures (i.e., ResNet50 or VGG16 or MobileNet), to create the three architectures. The rest of those proposed architectures; includes a feature extraction with attention combination; which is represented by a Bidirectional layer that covers a ConvLSTM layer with 16 filters,  $2 \times 2$  kernel size, and ReLU activation function; then a CBAM is added; followed by a MaxPooling3D layer with  $1 \times 2 \times 2$  pool size. The last

part of those architectures contains three layers, to take the final decision of video type (i.e., violence or non-violence); a Flatten layer inside a TimeDistributed layer is added; then the LSTM layer with 32 units; and the last decision layer of Dense layer with Softmax activation.

## III. DATA

Considering the existing violence detection data sets and the proposed work, the conclusion came to the necessity of a large-scale dataset, to apply the proposed work techniques. Unfortunately, most of the existing data sets suffer from drawbacks such as very low size, unlabeled videos, and a very limited number of scenarios. Based on these factors, the UBI-Fights data is used as a reference for the performance evaluation of the proposed work with respect to the most recent paper on the same data. This data is existed in [28], which has a variety of the violence scenarios from different perspectives (i.e., indoor or outdoor environments, the video's colors, the scene angles, the moving or stationary camera, and the degree of camera rotation). This data is collected and annotated based on frame levels (The videos were obtained from YouTube and LiveLeak using a wide range of text search results for violent events.), to form an 80 hour of recording; which is represented in 1000 videos (216 videos for different violence scenarios, and 784 videos for non-violence situations).

As a data preprocessing, there are two challenges in this data: 1) Categories or class imbalance, and 2) Memory limitations contain during model training on videos. Firstly, to solve the category imbalance challenge, the violence category is augmented by using different transformations (i.e., rotation, flipping, and noise), to be in the same range as the non-violence videos category (the number of videos for the violence category after augmentation is 864 videos). For the second challenge of memory limitation constraints, each video in the data is divided into 25 frames only, which covers the overall video. So, the proposed algorithm should take the decision of each video category by using this number of frames only.

## IV. EXPERIMENTS

In this part, the training process is summarized, and an ablation research is proposed to evaluate the value of proposed architectures based on the attention mechanism (i.e. CBAM). To assess the contribution and innovative results of this paper, a meaningful comparison between the proposed work and the state-of-the-art work on the same dataset (i.e., UBI-Fights) is also presented.

### A. TRAINING METHODOLOGY

For the UBI-Fights data preparation, as said in the data section, each video is represented by 25 frames only, distributed on overall video time interval. This technique proves its effectiveness to save memory size, without reducing the overall performance of the proposed classification models. In addition, the data augmentation for the low sampled category



(i.e., violence videos); by using different methods like flipping, rotating, and noise adding; to increase the violence videos to 864 videos, while the non-violence videos are 784. Then, each frame pixels are normalized with respect to 255 number, to get pixels' values range of 0 to 1. In the final preparation step before feeding into the models, the data is shuffled and spitted into training, validating, and testing sets (with percentages of 70, 15, and 15% respectively, based on the author's experience [29], [30]).

All proposed architectures are implemented and tested by using the Tensorflow library in the Python programming language; and the used computer has a Windows 10 operating system, and hardware specifications of (12th Gen Intel(R) Core(TM) i7-12700K 3.60 GHz) processor, (32.0 GB) RAM size, and (NVIDIA GeForce RTX 3090) GPU. During training, the number of videos per batch is 50 video, to get the highest possible performance of each model (the low number of video batches doesn't fulfill convergence during training, or tends to a fluctuated result; so the high batch value is better in such cases). To attend the convergence, the Adams optimizer is selected as the optimization method, with the default configuration parameters, through all training processes. Furthermore, the use of categorical focal loss as the loss function during training; to overcome the class imbalance problem in the raw data, and increase the focus on the main violence videos features. Finally, the early stopping technique based on the validation data loss value, is used to prevent the overfitting problems.

## B. EVALUATION METRICS

For the violence identification case study, the common evaluation metrics that used to evaluate the different models' performance, are the Area Under the Curve (AUC), Equal Error Rate (EER), and accuracy metrics. The AUC reflects the degree or measure of categories separability by the trained model, making it one of the most crucial assessment metrics for assessing the effectiveness of any classification model. An excellent model has an AUC close to 1, indicating that it has a high level of separability. In other words, the more accurate the model is in classifying non-violence classes as non-violence and violence classes as violence, the higher the AUC.

While the second evaluation metric, the EER by definition, refers to a kind of error margin that uses an algorithm to balance the False Rejections Rate (FRR) and False Acceptance Rate (FAR). In other words, an ERR is a point where FAR and FRR intersect or be equal; that is, the proportion of false acceptances is equal to the proportion of false rejections. This point serves as a performance indicator for violence detection models. The accuracy of the violence detection model increases with a decrease in the equal error rate value, because the false classification probability is too small.

Finally, the accuracy metric represents the total number of videos that are correctly classified, divided by the overall number of videos. This metric only is not enough because it

doesn't give a deep understanding of the classification model performance. So, it is not commonly used in the previous work on UBI-Fights data. Herby, to evaluate the performance of the proposed work, the results are mainly based on AUC, and EER values for each model with respect to test (unseen) data, which is used to make the ablation study and state-of-the-art comparisons.

## C. ABLATION STUDY

To make the ablation study, six different architectures based on the attention technique (CBAM), are proposed and evaluated by using the unseen UBI-Fights dataset. Those six proposed architectures are divided into three simple architectures created from scratch based on integrating the CBAM with ConvLSTM2D or Conv2D&LSTM. While the other three architectures are based on the transfer learning technique, by using the CBAM at the end of ready models like ResNet50 or VGG16 or MobileNet. The results of this ablation study indicate the ability for the first three simple architectures to get high violence detection performance, which can almost be similar to the very complicated architectures (i.e., the last three ones that are based on ready models and transfer learning). In addition, most of the proposed architectures can achieve very high AUC values, and very low EER values (with respect to the state-of-the-art results on the same UBI-Fights data).

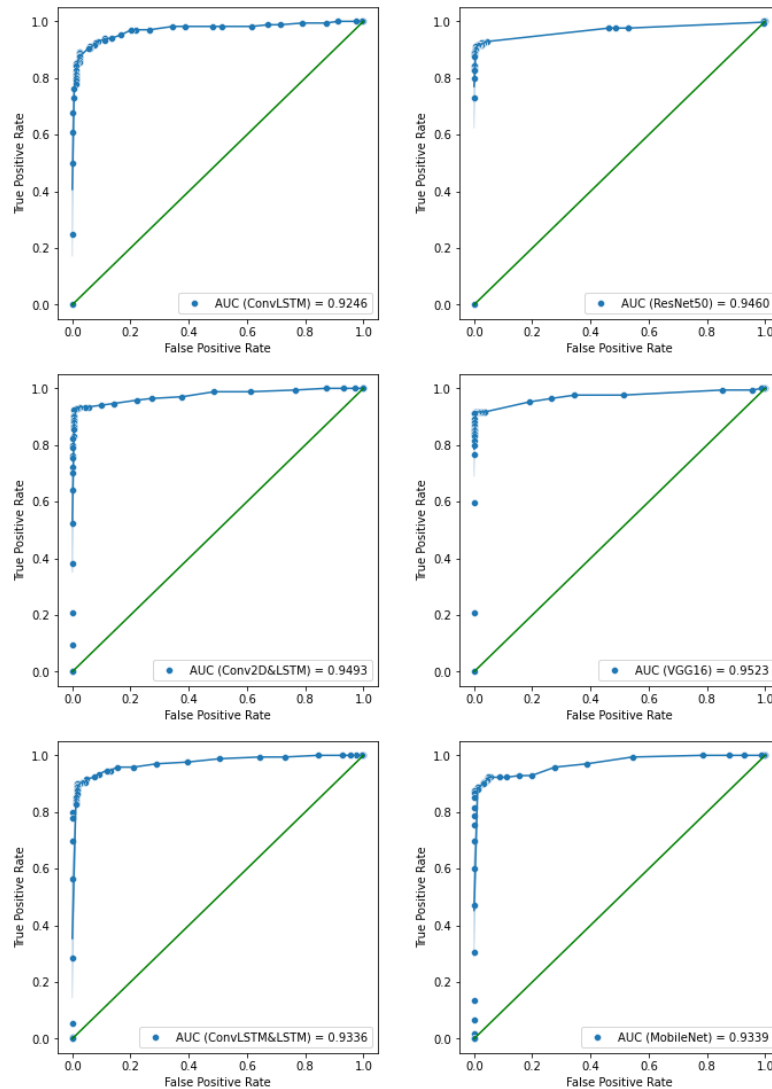
## V. RESULTS AND ANALYSIS

In this section, the results and analysis of this paper are summarized in two subsections. The first one is the ablation study results, which propose the evaluation results for the proposed architectures and compare these results. The second subsection is the state-of-the-art comparison, which presents a clear comparison between the proposed architectures and the state-of-the-art results on the UBI-Fights data, by using the AUC and EER evaluation metrics.

### A. ABLATION STUDY RESULTS

The ablation study focuses on the ability of simple architectures that created from scratch by integrating the CBAM with ConvLSTM2D or Conv2D layers; to achieve high violence detection performance that results from complicated architectures based on ready models as a backbone (e.g., ResNet50, VGG16, and MobileNet). All models are evaluated by using the accuracy, categorical focal loss value, AUC, and ERR (as a comparison criterion for the models' performance); and applying it to the UBI-Fights unseen part, to be sure about the models' performance in real life. As shown in Figure 4, the proposed simple architectures (i.e., ConvLSTM-based architecture, Conv2D&LSTM-based architecture, and ConvLSTM2D&LSTM-based architecture) achieve very high AUC results of 0.9246, 0.9493, and 0.9336 respectively; that represent a very high performance with respect to the complicated architectures (i.e., ResNet-based architecture, VGG-based architecture, and MobileNet-based architecture) that get AUC values of 0.9460, 0.9523, and





**FIGURE 4.** The ROC (receiver operating characteristic) curve for the all tested architectures by using UBI-Fights data, to demonstrate the nature of AUC value of different models (the simple architectures achieves high performance as well as complicated ones).

9339 respectively. For more declaration, it is clear that some simple architectures like Conv2D&LSTM-based architecture can achieve evaluation results higher than some complicated architectures like ResNet-based, and MobileNet-based architectures; that means saving in computation cost and fast response time in real life, with significant performance. Also, the ConvLSTM-based, and ConvLSTM2D&LSTM-based architectures, can get high-performance results, with still a little bit lower than the other ones.

For more ablation results declaration, Table 2 contain the details of each proposed architecture performance. It is clear that the accuracy metric only is not suitable for evaluating the best model performance; because some architectures like Conv2D&LSTM-based architectures get a high accuracy value of 0.9545 (higher than the VGG-based architecture that gets 0.9515 accuracy). However, the actual performance of the Conv2D&LSTM-based architecture,

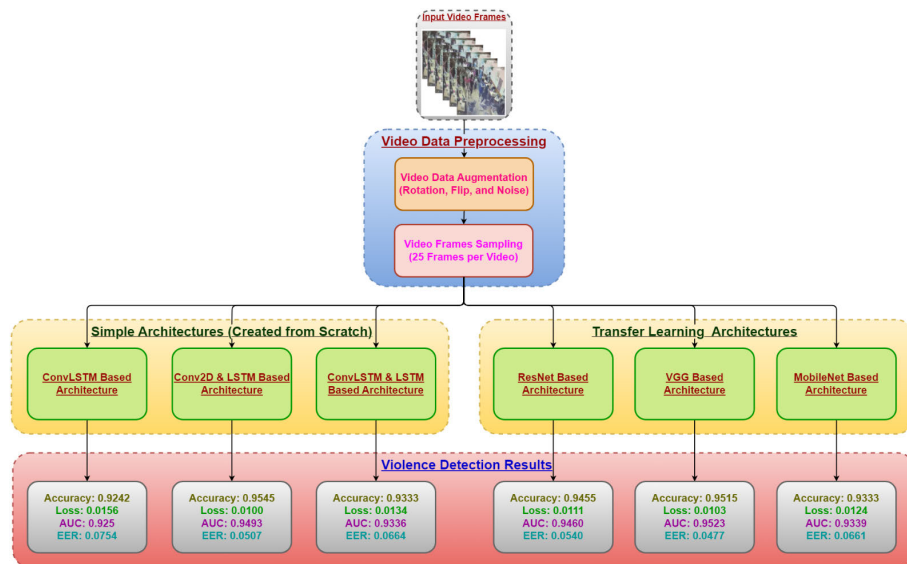
is declared by the 0.9493 AUC, and 0.0507 EER values, that lower than the AUC of 0.9523, and EER of 0.0477 for the VGG-based architecture. In other words, the actual Conv2D&LSTM-based architecture performance in real life, is lower than the VGG-based architecture performance. Because the ability to distinguish between violence and non-violence classes is lower than the VGG-based architecture; and the interaction or misclassification of the two categories, is slightly lower. Hereby, in the state-of-the-art comparison subsection, the main comparison criteria will be the AUC and EER values. Furthermore, Figure 5 demonstrates the overall work layout, and the results of each architecture.

## B. STATE-OF-THE-ART COMPARISON

To declare the contribution, novelty, and value of the proposed work through this paper; an additional performance comparison for the simply proposed architectures

**TABLE 2.** The performance results comparison of the all proposed architectures by using UBI-Fights data, in terms of accuracy, categorical focal loss value, AUC, and EER.

| Architecture                                                     | Accuracy      | Loss          | AUC           | EER           |
|------------------------------------------------------------------|---------------|---------------|---------------|---------------|
| ConvLSTM2D+CBAM<br>(ConvLSTM-Based Architectures)                | 0.9242        | 0.0156        | 0.925         | 0.0754        |
| <b>Conv2D&amp;LSTM+CBAM</b><br>(Conv2D&LSTM-Based Architectures) | <b>0.9545</b> | <b>0.0100</b> | <b>0.9493</b> | <b>0.0507</b> |
| ConvLSTM2D&LSTM+CBAM<br>(ConvLSTM2D&LSTM-Based Architectures)    | 0.9333        | 0.0134        | 0.9336        | 0.0664        |
| ResNet50+CBAM<br>(ResNet-Based Architectures)                    | 0.9455        | 0.0111        | 0.9460        | 0.0540        |
| <b>VGG16+CBAM</b><br>(VGG-Based Architectures)                   | <b>0.9515</b> | <b>0.0103</b> | <b>0.9523</b> | <b>0.0477</b> |
| MobileNet+CBAM<br>(MobileNet-Based Architectures)                | 0.9333        | 0.0124        | 0.9339        | 0.0661        |

**FIGURE 5.** The overall work layout, including video data preprocessing and a full results declaration.

(i.e., ConvLSTM-based architecture, Conv2D&LSTM-based architecture, and ConvLSTM2D&LSTM-based architecture) with respect to the state-of-the-art techniques that used to detect violence in surveillance videos, is added (Table 3). This comparison declares the significant performance of the proposed architectures (especially Conv2D&LSTM-based architecture), with respect to the state-of-the-art models. From the comparison table (Table 3), it is clear that most of the proposed work is outperform the state-of-the-art results. For example, the Conv2D&LSTM-based architecture got a 0.9493 AUC value, that higher than the highest one in the existing papers (i.e., an AUC value of 0.931 for the two-stage weakly supervised framework in [8]) for the same UBI-Fights data. Also, the EER value for the same proposed architecture

is lower than the state-of-the-art values (i.e., much higher classification performance with a much more simple model).

## VI. DISCUSSION

The proposed architectures proof its ability to detect violence in surveillance videos by extracting the spatio-temporal features. This is explained through the previous section (i.e., results and analysis section). The high-performance results, are achieved by creating some simple architectures that are based on ConvLSTM2D layer or Conv2D&LSTM layers; and the UBI-Fights dataset is used as a case study, to evaluate the proposed work by comparing to the existing papers' results. Due to the integration of CBAM with those simple architectures, the spatio-temporal feature extraction

**TABLE 3.** The performance results comparison of the proposed simple architectures, with respect to the most state-of-the-art techniques on the same UBI-Fights data, in terms of AUC, and EER.

| Model                                                                              | AUC           | EER           |
|------------------------------------------------------------------------------------|---------------|---------------|
| Weakly and Partially Supervised Iterative Learning Framework [18]                  | 0.819         | 0.284         |
| Combine R(2+1)D architecture with BERT attention [19]                              | 0.915         | —             |
| XG-Boost Classifier with Moving Human Identification Preprocessing [21]            | 0.769         | 0.427         |
| <b>Weakly Supervised training of Two Stage Framework [8]</b>                       | <b>0.931</b>  | <b>0.085</b>  |
| ConvLSTM2D+CBAM<br>(ConvLSTM <sub>2D</sub> -Based Architectures)                   | 0.925         | 0.0754        |
| <b>Conv2D&amp;LSTM+CBAM<br/>(Conv2D&amp;LSTM<sub>2D</sub>-Based Architectures)</b> | <b>0.9493</b> | <b>0.0507</b> |
| ConvLSTM2D&LSTM+CBAM<br>(ConvLSTM2D&LSTM <sub>2D</sub> -Based Architectures)       | 0.9336        | 0.0664        |

process is possible, even with the challenges of different videos quality, different environments, and various camera perspectives (e.g., rotation, and movements). Furthermore, the Conv2D&LSTM-based architecture outperforms the results of state-of-the-art methods. At the same time, get a performance slightly lower than the complicated architectures like VGG-based one; and higher than some others, that used ready models (e.g., ResNet50, and MobileNet) as a backbone.

## VII. CONCLUSION AND FUTURE WORK

In this paper, the violence detection in surveillance videos task is comprehensively studied by using UBI-Fights as a reference dataset, to evaluate the proposed work. This study is done through steps: 1) Presenting a clear and deep demonstration of the problem case study and challenges, by making a review for the most recent related work on the same data; 2) Creating the different architectures that fulfill the requirements, that identified through the first step; and 3) Evaluating the proposed work by comparing to each other, and comparing to the state-of-the-art work.

Six different architectures, are implemented and evaluated by using the UBI-Fights dataset. Those architectures contain three simple ones, that were created from scratch by integrating the Convolutional Block Attention Module (CBAM) with the ConvLSTM2D or Conv2D&LSTM layers; and other three ones have a similar integration process based on ResNet50 or VGG16 or MobileNet, as a spatio-temporal feature extractor. The CBAM includes two main attention modules type called the channel attention module, and the spatial attention module; they both are constructed sequentially to increase the architectures' attention to the most important features like the

nature of people's interactions, and neglect the insufficient ones like the environment features. In addition, the use of Categorical Focal Loss (CFL) as a loss function through the architectures' training, reduces the imbalance data problem drawbacks, and increases the models' focus on the most important features. The evaluation criteria or metrics are mainly based on the Area Under the Curve (AUC), and Equal Error Rate (EER) metrics. Also, the evaluation is done based on two main comparisons: 1) Ablation study Comparison that highlights the significant performance of the simple proposed architectures with respect to the other complicated ones; and 2) State-of-the-art comparison that declare the innovation performance of the proposed work with respect to the existing papers on the UBI-Fights data.

The comparisons steps for performance results demonstrate the ability of the Conv2d&LSTM-based architecture; to achieve a high performance of 0.9493 for the AUC metric and 0.0507 for the EER metric. These values indicate the architecture's ability to distinguish between categories with low interaction probabilities. At the same time, the performance of this architecture is almost similar to the complicated architectures' performance; which means saving computation cost and memory size, with the same resultant performance. Also, the most of proposed architectures, achieve high performance by comparing to state-of-the-art performance. Furthermore, the importance of using evaluation metrics like AUC, and EER in such cases, instead of accuracy metric only, is demonstrated by values through the comparisons.

Finally, the future work for this paper will extend the proposed work to various data, to achieve a generalization of violence detection tasks. Also, integrating the proposed work with other techniques (e.g., optical flow or pose estimation or

semantic segmentation or object detection) is should be done, to identify the limitation of this task.

## ACKNOWLEDGMENT

This research was in part supported by the MSIT (Ministry of Science and ICT), Korea, under the Grand Information Technology Research Center support program (IITP-2023-2020-0-01462) supervised by the IITP (Institute for Information & communications Technology Planning & Evaluation), and in part by the Research Projects of “Development of automatic screening and hybrid detection system for hazardous material detecting in port container” funded by the Ministry of Oceans and Fisheries.

## REFERENCES

- [1] S. Davila-Montero, J. A. Dana-Le, G. Bente, A. T. Hall, and A. J. Mason, “Review and challenges of technologies for real-time human behavior monitoring,” *IEEE Trans. Biomed. Circuits Syst.*, vol. 15, no. 1, pp. 2–28, Feb. 2021.
- [2] B. Fan, P. Li, S. Jin, and Z. Wang, “Anomaly detection based on pose estimation and GRU-FFN,” in *Proc. IEEE Sustain. Power Energy Conf. (iSPEC)*, Dec. 2021, pp. 3821–3825.
- [3] B. Cao, H. Xia, and Z. Liu, “A video abnormal behavior recognition algorithm based on deep learning,” in *Proc. IEEE 4th Adv. Inf. Manage., Communicates, Electron. Autom. Control Conf. (IMCEC)*, vol. 4, Jun. 2021, pp. 755–759.
- [4] R. Vrskova, R. Hudec, P. Kamencay, and P. Sykora, “Recognition of human activity and abnormal behavior using deep neural network,” in *Proc. ELEKTRO (ELEKTRO)*, May 2022, pp. 1–4.
- [5] F. J. Rendón-Segador, J. A. Álvarez-García, F. Enríquez, and O. Deniz, “ViolenceNet: Dense multi-head self-attention with bidirectional convolutional LSTM for detecting violence,” *Electronics*, vol. 10, no. 13, p. 1601, Jul. 2021.
- [6] S. Sharma, B. Sudharsan, S. Narahariseti, V. Trehan, and K. Jayavel, “A fully integrated violence detection system using CNN and LSTM,” *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 11, no. 4, p. 3374, Aug. 2021.
- [7] A. Hanson, K. PNVR, S. Krishnagopal, and L. Davis, “Bidirectional convolutional LSTM for the detection of violence in videos,” presented at the Eur. Conf. Comput. Vis. (ECCV), Munich, Germany, Sep. 2018.
- [8] Z. Qi, R. Zhu, Z. Fu, W. Chai, and V. Kindratenko, “Weakly supervised two-stage training scheme for deep video fight detection model,” presented at the Int. Conf. Tools With Artif. Intell. (ICTAI), Sep. 2022.
- [9] Y. S. Chong and Y. H. Tay, “Abnormal event detection in videos using spatiotemporal autoencoder,” in *Proc. Int. Symp. Neural Netw. (ISNN)*, vol. 10262, 2017, pp. 189–196.
- [10] M. Hasan, J. Choi, J. Neumann, A. K. Roy-Chowdhury, and L. S. Davis, “Learning temporal regularity in video sequences,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 733–742.
- [11] M. Sabokrou, M. Fayyaz, M. Fathy, and R. Klette, “Deep-cascade: Cascading 3D deep neural networks for fast anomaly detection and localization in crowded scenes,” *IEEE Trans. Image Process.*, vol. 26, no. 4, pp. 1992–2004, Apr. 2017.
- [12] W. Shin, S.-J. Bu, and S.-B. Cho, “3D-convolutional neural network with generative adversarial network and autoencoder for robust anomaly detection in video surveillance,” *Int. J. Neural Syst.*, vol. 30, no. 6, Jun. 2020, Art. no. 2050034.
- [13] S. Zhou, W. Shen, D. Zeng, M. Fang, Y. Wei, and Z. Zhang, “Spatial-temporal convolutional neural networks for anomaly detection and localization in crowded scenes,” *Signal Process., Image Commun.*, vol. 47, pp. 358–368, Sep. 2016.
- [14] J. R. Medel and A. Savakis, “Anomaly detection in video using predictive convolutional long short-term memory networks,” 2016, *arXiv:1612.00390*.
- [15] S. Sudhakaran and O. Lanz, “Learning to detect violent videos using convolutional long short-term memory,” in *Proc. 14th IEEE Int. Conf. Adv. Video Signal Based Surveill. (AVSS)*, Sep. 2017, pp. 1–6.
- [16] Ş. Aktı, G. A. Tataroğlu, and H. K. Ekenel, “Vision-based fight detection from surveillance cameras,” in *Proc. 9th Int. Conf. Image Process. Theory, Tools Appl. (IPTA)*, Nov. 2019, pp. 1–6.
- [17] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “CBAM: Convolutional block attention module,” presented at the Eur. Conf. Comput. Vis. (ECCV), Munich, Germany, Sep. 2018.
- [18] B. M. Degardin, “Weakly and partially supervised learning frameworks for anomaly detection,” M.S. thesis, Comput. Eng., Universidade da Beira Interior (UBI), Covilha, Portugal, 2020.
- [19] W. Tan and J. Liu, “Detection of fights in videos: A comparison study of anomaly detection and action recognition,” 2022, *arXiv:2205.11394*.
- [20] M. E. Kalfaoglu, S. Kalkan, and A. A. Alatan, “Late temporal modeling in 3D CNN architectures with BERT for action recognition,” in *Proc. Eur. Conf. Comput. Vis. Cham, Switzerland: Springer*, 2020, pp. 731–747.
- [21] M. Alarfaj, M. Pervaiz, Y. Y. Ghadi, T. al Shloul, S. A. Alsuhibany, A. Jalal, and J. Park, “Automatic anomaly monitoring in public surveillance areas,” *Intell. Autom. Soft Comput.*, vol. 35, no. 3, pp. 2655–2671, 2023.
- [22] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
- [23] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.
- [24] L. Chen, H. Zhang, J. Xiao, L. Nie, J. Shao, W. Liu, and T.-S. Chua, “SCA-CNN: Spatial and channel-wise attention in convolutional networks for image captioning,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6298–6306.
- [25] T. Liu, R. Luo, L. Xu, D. Feng, L. Cao, S. Liu, and J. Guo, “Spatial channel attention for deep convolutional neural networks,” *Mathematics*, vol. 10, no. 10, p. 1750, May 2022.
- [26] T. Shan and J. Yan, “SCA-Net: A spatial and channel attention network for medical image segmentation,” *IEEE Access*, vol. 9, pp. 160926–160937, 2021.
- [27] M. Kerkhof, L. Wu, G. Perin, and S. Picek, “Focus is key to success: A focal loss function for deep learning-based side-channel analysis,” in *Constructive Side-Channel Analysis and Secure Design*. Cham, Switzerland: Springer, 2022, pp. 29–48.
- [28] B. Degardin and H. Proença, “Human activity analysis: Iterative weak/self-supervised learning frameworks for detecting abnormal events,” in *Proc. IEEE Int. Joint Conf. Biometrics (IJCB)*, Oct. 2020, pp. 1–7.
- [29] M. A. B. Abbass and M. Hamdy, “A generic pipeline for machine learning users in energy and buildings domain,” *Energies*, vol. 14, no. 17, p. 30, 2021.
- [30] M. A. B. Abbass, H. Sadek, and M. Hamdy, “Buildings energy prediction using artificial neural networks,” *Eng. Res. J. (EJR)*, vol. 171, p. 12, Sep. 2021.



ing, applied machine learning, computer vision, and deep learning.



of Electrical and Computer Engineering, Chungbuk National University, Chungbuk, Republic of Korea, in March 2005. His research interests include image compression and image processing.

• • •